Operation Management

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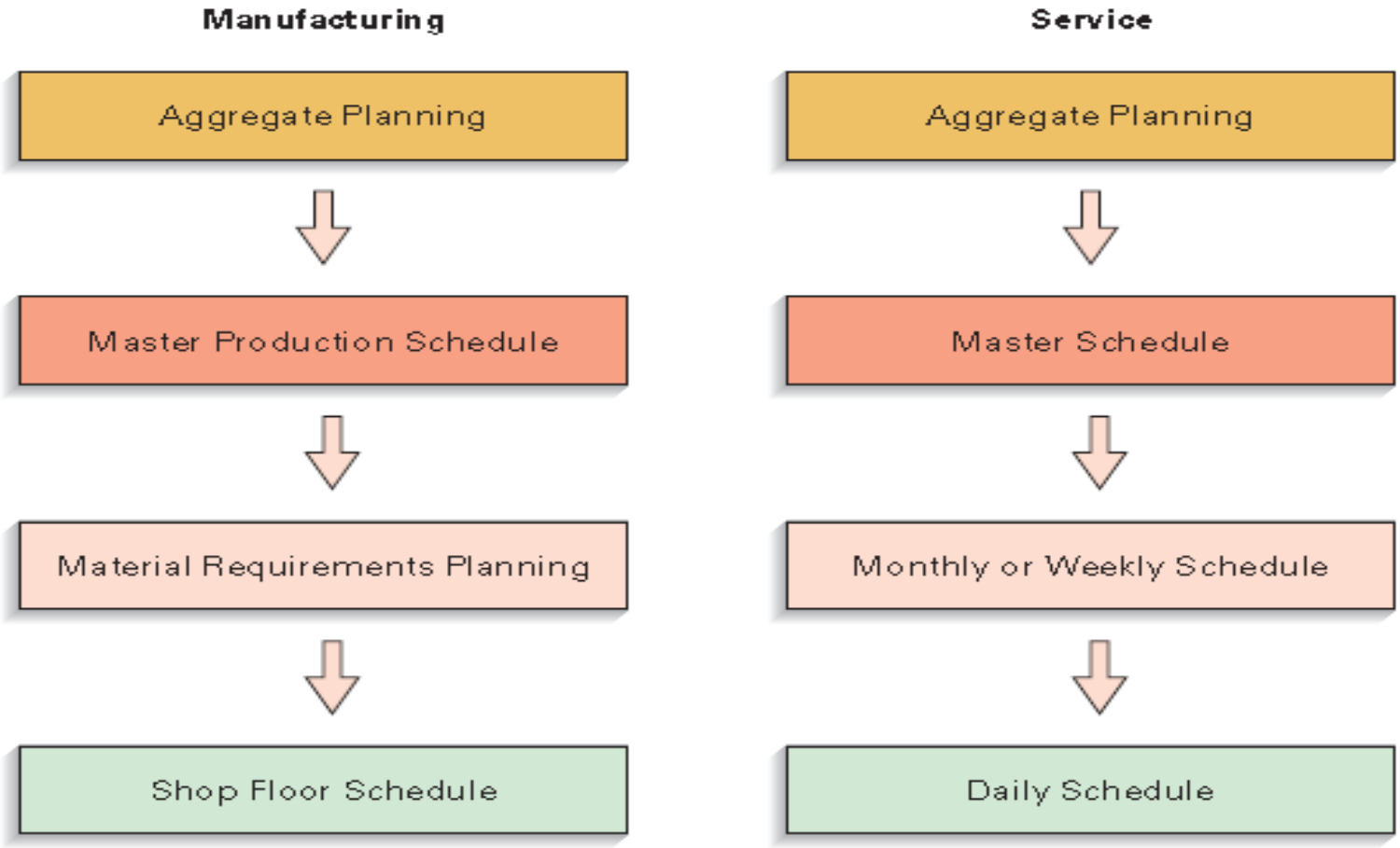
<MBAZGC526, Operation Management>
Lecture No. 12

Planning Hierarchy

- Capacity Planning.
- Aggregate Planning
- Master Production Schedule.
- Material Requirement Planning.
- Short Scheduling.

Planning Hierarchy

Prepared by Prof. Abhishek Maiti for the use of the students of Operation Management



Source: Operation Management by William J Stevenson

Scheduling

- Establishing the timing of the use of equipment, facilities, and human activities in an organization.
- The objective of scheduling is to allocate and prioritize demand to available facilities.
- Factors are pervasive in scheduling: (1) Forward and Backward Scheduling (2) Finite and Infinite loading, and (3) Criteria (priorities) for sequencing jobs.

Importance of Short-Term Scheduling

- **Effective and efficient scheduling can be a competitive advantage**
  - Faster movement of goods through a facility means better use of assets and lower costs
  - Additional capacity resulting from faster throughput improves customer service through faster delivery
  - Good schedules result in more dependable deliveries

Forward Scheduling

- Forward scheduling starts the schedule *as soon as the job requirements are known* .
- Hospitals, restaurants, and machine tool manufacturers.
- Jobs are performed to customer order.

Backward Scheduling

- Backward scheduling *begins with the due date*, scheduling the *final* operation first.
- Steps within the job are then scheduled, one at a time, in reverse order.
- By subtracting the time needed for each item, the start time is obtained.
- Scheduling surgery.

Why Scheduling

- Minimize completion time.
- Maximize utilization.
- Minimize work-in-process (WIP) inventory.
- Minimize customer waiting time.

Scheduling Operations

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TABLE 15.2 Different Processes Suggest Different Approaches to Scheduling

Process-focused facilities (job shops)

- Scheduling to customer orders where changes in both volume and variety of jobs/clients/patients are frequent.
- Schedules are often due-date focused, with loading refined by finite loading techniques.
- *Examples:* foundries, machine shops, cabinet shops, print shops, many restaurants, and the fashion industry.

Repetitive facilities (assembly lines)

- Schedule module production and product assembly based on frequent forecasts.
- Finite loading with a focus on generating a forward-looking schedule.
- JIT techniques are used to schedule components that feed the assembly line.
- *Examples:* assembly lines for washing machines at Whirlpool and automobiles at Ford.

Product-focused facilities (continuous)

- Schedule high-volume finished products of limited variety to meet a reasonably stable demand within existing fixed capacity.
- Finite loading with a focus on generating a forward-looking schedule that can meet known setup and run times for the limited range of products.
- *Examples:* huge paper machines at International Paper, beer in a brewery at Anheuser-Busch, and potato chips at Frito-Lay.

Source: Heizer, Render, Muckson

Loading

- Loading is the process of assigning jobs to work stations or processes.
- Scheduling techniques that load (or assign) work only up to the capacity of the process are called *finite loading*.
- Scheduling techniques that load (or assign) work without considering the capacity of the process are *infinite loading* .

Loading Jobs

Two Forms:

- Capacity Oriented : Input-Output control.
- Assigning specific jobs to work centers : *Gantt charts* and the *assignment method*.

Input-Output Control

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Welding Work Center (In standard hours)

Week Ending	6/6	6/13	6/20	6/27	7/4	7/11
Planned Input	280	280	280	280	280	
Actual Input	270	250	280	285	280	
Cumulative Deviation	−10	−40	−40	−35		
Planned Output	320	320	320	320		
Actual Output	270	270	270	270		
Cumulative Deviation	−50	−100	−150	−200		
Cumulative Change in Backlog*	0	−20	−10	+5		

Explanation: 270 input, 270 output, implies 0 change.

Explanation: 250 input, 270 output, implies −20 change. (20 standard hours less work in the work center)

*Sum of actual inputs minus sum of actual outputs = cumulative change in backlog

Source: Heizer, Render, Muckson

ConWIP Cards

- Identifies overloading and underloading conditions
- Prompts managerial action to resolve scheduling problems
- Can be maintained using ConWIP cards that control the scheduling of batches

Gantt Load Chart

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Work Center \ Day	Monday	Tuesday	Wednesday	Thursday	Friday
Metalworks	Job 349		← Job 350 →		
Mechanical		← Job 349 →		Job 408	
Electronics	Job 408			Job 349	
Painting	← Job 295 →		Job 408		Job 349

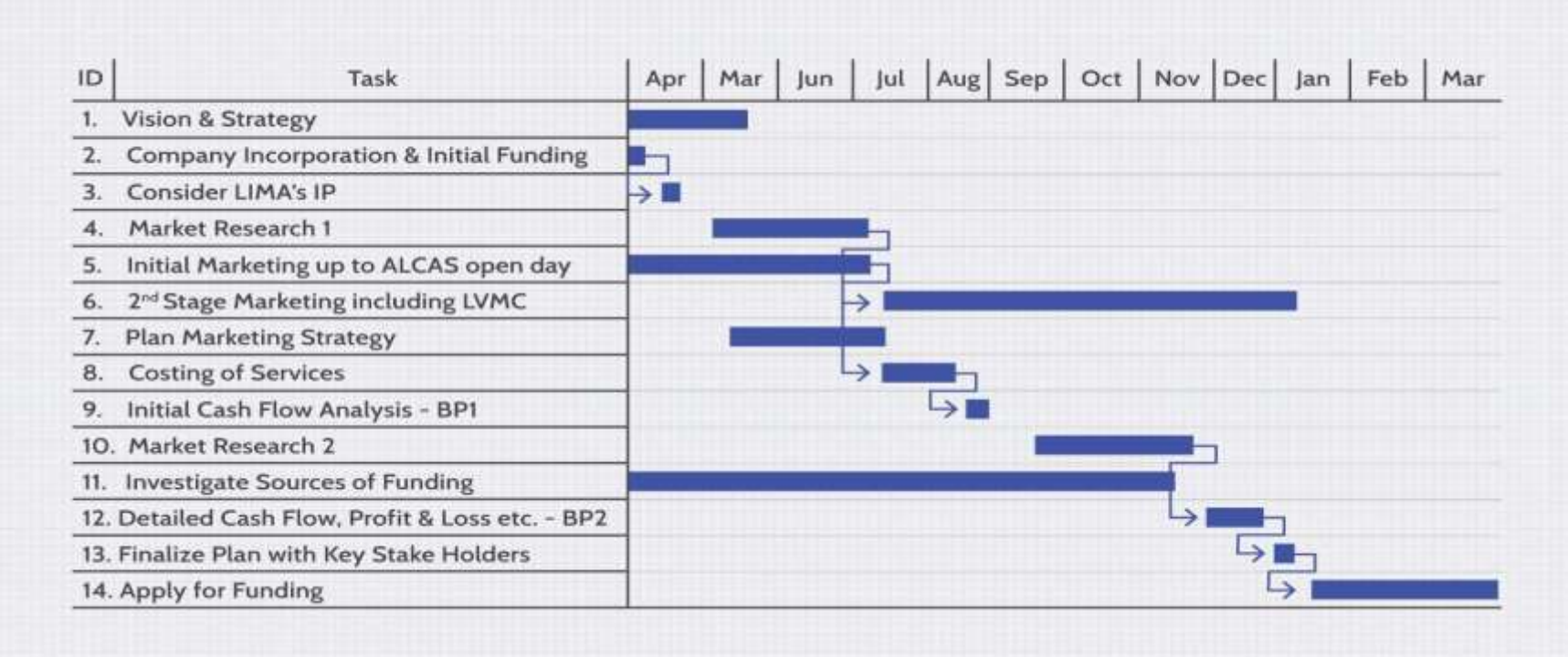
Processing | Unscheduled | Center not available (e.g., maintenance time, repairs, shortages)

- It does not account for production variability such as unexpected breakdowns or human errors.

Source: Heizer, Render, Muckson

Gantt Schedule Chart

- A Gantt *schedule chart* is used to monitor jobs in progress.



Assignment Method

- A special class of linear programming models that involves assigning tasks or jobs to resources.
- The objective is to minimize total costs or maximizing profit.
- One important characteristic of assignment problems is that only one job is assigned to one machine (or project).

Assignment Problem

- Cost Minimization.
- Revenue or Profit Maximization

Assignment Problem : Cost Minimization

Example 4

ASSIGNMENT METHOD

First Printing wants to find the minimum total cost assignment of 3 jobs to 3 typesetters.

APPROACH ▶ The cost table shown earlier in this section is repeated here, and steps 1 through 4 are applied.

JOB \ TYPESETTER	A	B	C
R-34	$11	$14	$ 6
S-66	$ 8	$10	$11
T-50	$ 9	$12	$ 7

STUDENT TIP

You can also tackle assignment problems with our Excel OM or POM software or with Excel's Solver add-in.

SOLUTION ▶

Source: Heizer, Render, Muckson

JOB \ TYPESETTER	A	B	C
R-34	5	8	0
S-66	0	2	3
T-50	2	5	0

JOB \ TYPESETTER	A	B	C
R-34	5	6	0
S-66	0	0	3
T-50	2	3	0

JOB \ TYPESETTER	A	B	C
R-34	5	6	0
S-66	0	0	3
T-50	2	3	0

Smallest uncovered number

JOB \ TYPESETTER	A	B	C
R-34	3	4	0
S-66	0	0	5
T-50	0	1	0

Return to step 2. Cover the zeros with straight lines again.

JOB \ TYPESETTER	A	B	C
R-34	3	4	0
S-66	0	0	
T-50	0	1	0

Source: Heizer, Render, Muckson

Class Exercise

- Scott and associates, Inc., is an accounting firm that has three new clients. Project leaders will be assigned to the three clients.
- Based on the different backgrounds and experiences of the leaders, the various leader client assignments differ in terms of project completion times.
- The possible assignments and the estimated completion times in days are as follows

Class Exercise

	Client		
Project Leader	1	2	3
Jackson	10	16	32
Ellis	14	22	40
Smith	22	24	34

Refer Excel

Any Question

Sequencing Jobs

- Sequencing : Determining the order in which jobs should be done at each work centre.

Priority rules for Sequencing Jobs

- **FCFS :** First come, first served.
- **SPT :** Shortest processing time.
- **EDD :** Earliest due date.
- **LPT :** Longest processing time.

Performance Criteria

$$\text{Average completion time} = \frac{\text{Sum of total flow time}}{\text{Number of jobs}} \tag{15-1}$$

$$\text{Utilization metric} = \frac{\text{Total job work (processing) time}}{\text{Sum of total flow time}} \tag{15-2}$$

$$\text{Average number of jobs in the system} = \frac{\text{Sum of total flow time}}{\text{Total job work (processing) time}} \tag{15-3}$$

$$\text{Average job lateness} = \frac{\text{Total late days}}{\text{Number of jobs}} \tag{15-4}$$

$$\text{Job lateness} = \text{Max}\{0, \text{yesterday} + \text{flow time} - \text{due date}\} \tag{15-5}$$

Source: Heizer, Render, Muckson

Class Exercise

- Five architectural rendering jobs are waiting to be assigned at Avanti Sethi Architects.
- Their work (processing) times and due dates are given in the table.
- The firm wants to determine the sequence of processing according to (1) FCFS, (2) SPT, (3) EDD, and (4) LPT rules.
- Jobs were assigned a letter in the order they arrived.
- Today is day 1, and work begins today.

Table

Job	Job Work (Processing) Time (Days)	Job Due Date (Days)
A	6	8
B	2	6
C	8	18
D	3	15
E	9	23

Source: Heizer, Render, Muckson

FCFS Sequence

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Job Sequence	Job Work (Processing) Time	Flow Time	Job Due Date	Job Lateness
A	6	6	8	0
B	2	8	6	2
C	8	16	18	0
D	3	19	15	4
E	9	28	23	5
	28	77		11

The FCFS rule results in the following measures of effectiveness:

a. Average completion time $= \dfrac{\text{Sum of total flow time}}{\text{Number of jobs}}$

$$= \frac{77 \text{ days}}{5} = 15.4 \text{ days}$$

b. Utilization metric $= \dfrac{\text{Total job work (processing) time}}{\text{Sum of total flow time}}$

$$= \frac{28}{77} = 36.4\%$$

c. Average number of jobs in the system $= \dfrac{\text{Sum of total flow time}}{\text{Total job work (processing) time}}$

$$= \frac{77 \text{ days}}{28 \text{ days}} = 2.75 \text{ jobs}$$

d. Average job lateness $= \dfrac{\text{Total late days}}{\text{Number of jobs}} = \dfrac{11}{5} = 2.2 \text{ days}$

Source: Heizer, Render, Muckson

SPT Sequence

Job Sequence	Job Work (Processing) Time	Flow Time	Job Due Date	Job Lateness
B	2	2	6	0
D	3	5	15	0
A	6	11	8	3
C	8	19	18	1
E	9	28	23	5
	28	65		9

Measurements of effectiveness for SPT are:

a. Average completion time $= \frac{65}{5} = 13$ days

b. Utilization metric $= \frac{28}{65} = 43.1\%$

c. Average number of jobs in the system $= \frac{65}{28} = 2.32$ jobs

d. Average job lateness $= \frac{9}{5} = 1.8$ days

Source: Heizer, Render, Muckson

EDD Sequence

3. The *EDD* rule shown in the next table gives the sequence B–A–D–C–E. Note that jobs are ordered by earliest due date first.

Job Sequence	Job Work (Processing) Time	Flow Time	Job Due Date	Job Lateness
B	2	2	6	0
A	6	8	8	0
D	3	11	15	0
C	8	19	18	1
E	9	28	23	5
	28	68		6

Measurements of effectiveness for EDD are:

a. Average completion time $= \frac{68}{5} = 13.6$ days

b. Utilization metric $= \frac{28}{68} = 41.2\%$

c. Average number of jobs in the system $= \frac{68}{28} = 2.43$ jobs

d. Average job lateness $= \frac{6}{5} = 1.2$ days

Source: Heizer, Render, Muckson

LPT Sequence

4. The *LPT* rule shown in the next table results in the order E–C–A–D–B.

Job Sequence	Job Work (Processing) Time	Flow Time	Job Due Date	Job Lateness
E	9	9	23	0
C	8	17	18	0
A	6	23	8	15
D	3	26	15	11
B	2	28	6	22
	28	103		48

Measures of effectiveness for LPT are:

a. Average completion time $= \frac{103}{5} = 20.6$ days

b. Utilization metric $= \frac{28}{103} = 27.2\%$

c. Average number of jobs in the system $= \frac{103}{28} = 3.68$ jobs

d. Average job lateness $= \frac{48}{5} = 9.6$ days

Source: Heizer, Render, Muckson

Summary

Rule	Average Completion Time (days)	Utilization Metric (%)	Average Number of Jobs in System	Average Lateness (days)
FCFS	15.4	36.4	2.75	2.2
SPT	13.0	43.1	2.32	1.8
EDD	13.6	41.2	2.43	1.2
LPT	20.6	27.2	3.68	9.6

Source: Heizer, Render, Muckson

Any Question

Critical Ratio

$$CR = \frac{\text{Time remaining}}{\text{Workdays remaining}} = \frac{\text{Due date} - \text{Today's date}}{\text{Work (lead) time remaining}} \qquad (15\text{-}6)$$

Class Exercise

- Today is day 25 on Zyco Medical Testing Laboratories' production schedule. Three jobs are on order, as indicated here:

JOB	DUE DATE	WORKDAYS REMAINING
A	30	4
B	28	5
C	27	2

Source: Heizer, Render, Muckson

Solution

JOB	CRITICAL RATIO	PRIORITY ORDER
A	(30 − 25)/4 = 1.25	3
B	(28 − 25)/5 = .60	1
C	(27 − 25)/2 = 1.00	2

- Job B has a critical ratio of less than 1, meaning it will be late unless expedited. Thus, it has the highest priority.
- Job C is on time, and job A has some slack.
- Once job B has been completed, we would re-compute the critical ratios for jobs A and C to determine whether their priorities have changed.

Source: Heizer, Render, Muckson

Sequencing N Jobs on Two Machines

- The next step in complexity is the case in which N jobs (where N is 2 or more) must go through two different machines or work centers in the same order.
- This is called the $N/2$ problem.
- **Johnson's rule** : An approach that minimizes the total time for sequencing a group of jobs through two work centers while minimizing total idle time in the work centres.

Class Exercise

- Five specialty jobs at a La Crosse, Wisconsin, tool and die shop must be processed through two work centers (drill press and lathe).
- The time for processing each job follows. The owner, Niranjan Pati, wants to set the sequence to minimize his total time for the five jobs

Work (processing) Time for Jobs (hours)

JOB	WORK CENTER 1 (DRILL PRESS)	WORK CENTER 2 (LATHE)
A	5	2
B	3	6
C	8	4
D	10	7
E	7	12

Source: Heizer, Render, Muckson

Solution

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SOLUTION ►

1. The job with the shortest processing time is A, in work center 2 (with a time of 2 hours). Because it is at the second center, schedule A last. Eliminate it from consideration.

				A

2. Job B has the next shortest time (3 hours). Because that time is at the first work center, we schedule it first and eliminate it from consideration.

B				A

3. The next shortest time is job C (4 hours) on the second machine. Therefore, it is placed as late as possible.

B			C	A

4. There is a tie (at 7 hours) for the shortest remaining job. We can place E, which was on the first work center, first. Then D is placed in the last sequencing position:

B	E	D	C	A

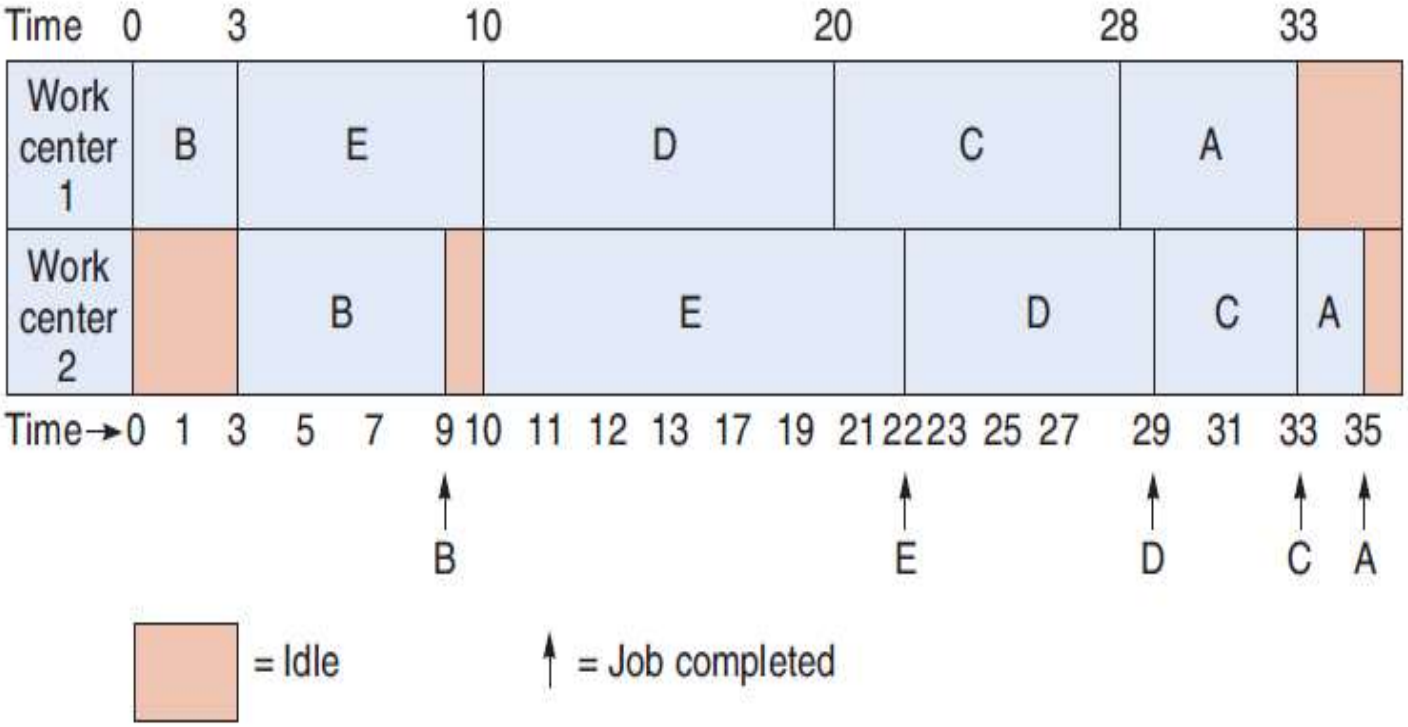
The sequential times are:

Work center 1	3	7	10	8	5
Work center 2	6	12	7	4	2

Source: Heizer, Render, Muckson

Solution

The time-phased flow of this job sequence is best illustrated graphically:



Thus, the five jobs are completed in 35 hours.

Source: Heizer, Render, Muckson

Limitations of Rule-Based Dispatching Systems

1. Scheduling is dynamic and rules need to be revised to adjust to changes
2. Rules do not look upstream or downstream
3. Rules do not look beyond due dates

Finite Capacity Scheduling

- Overcomes disadvantages of rule-based systems by providing an interactive, computer-based graphical system
- May include rules and expert systems or simulation to allow real-time response to system changes
- FCS allows the balancing of delivery needs and efficiency

Finite Capacity Scheduling

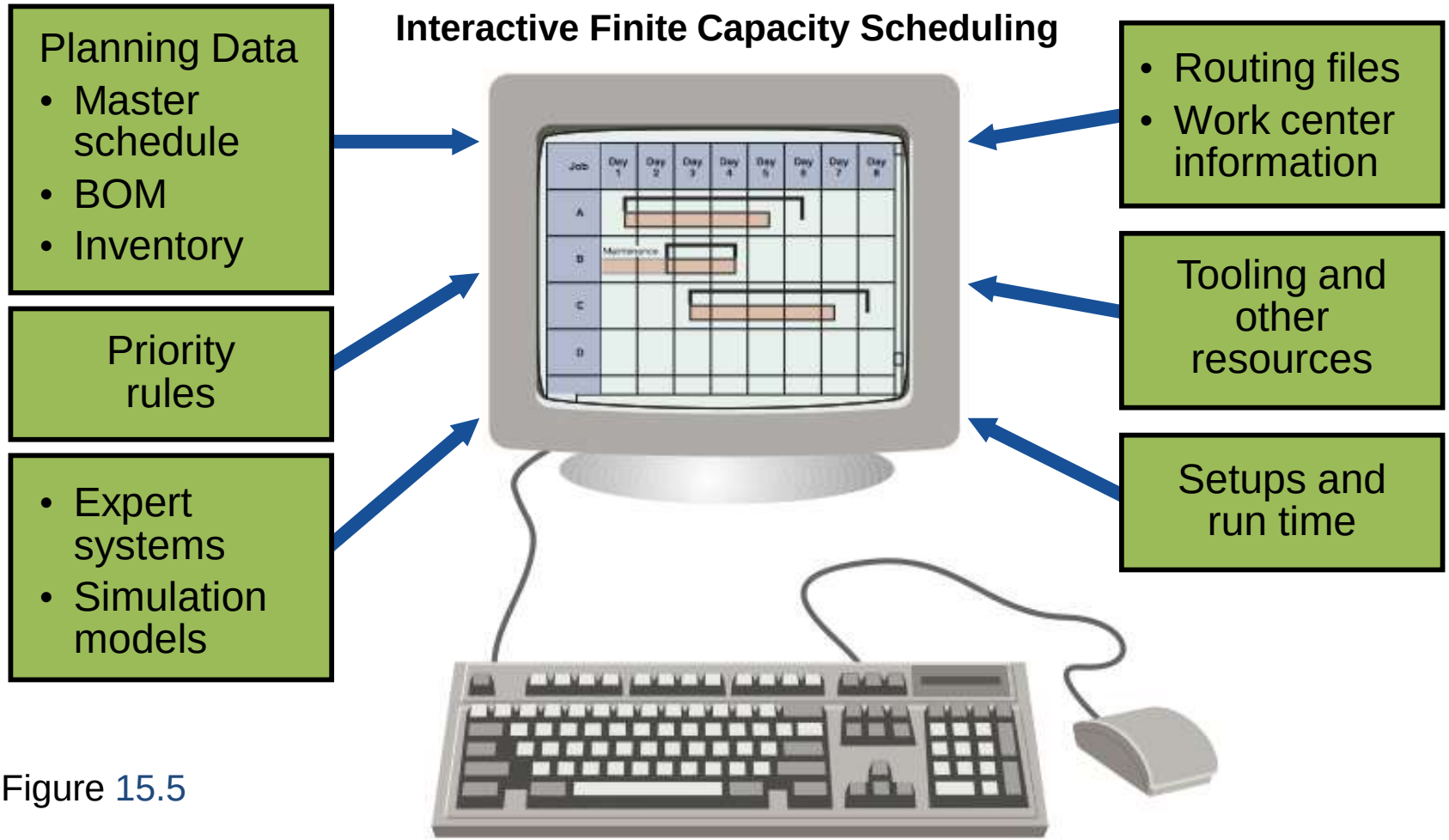


Figure 15.5

Finite Capacity Scheduling

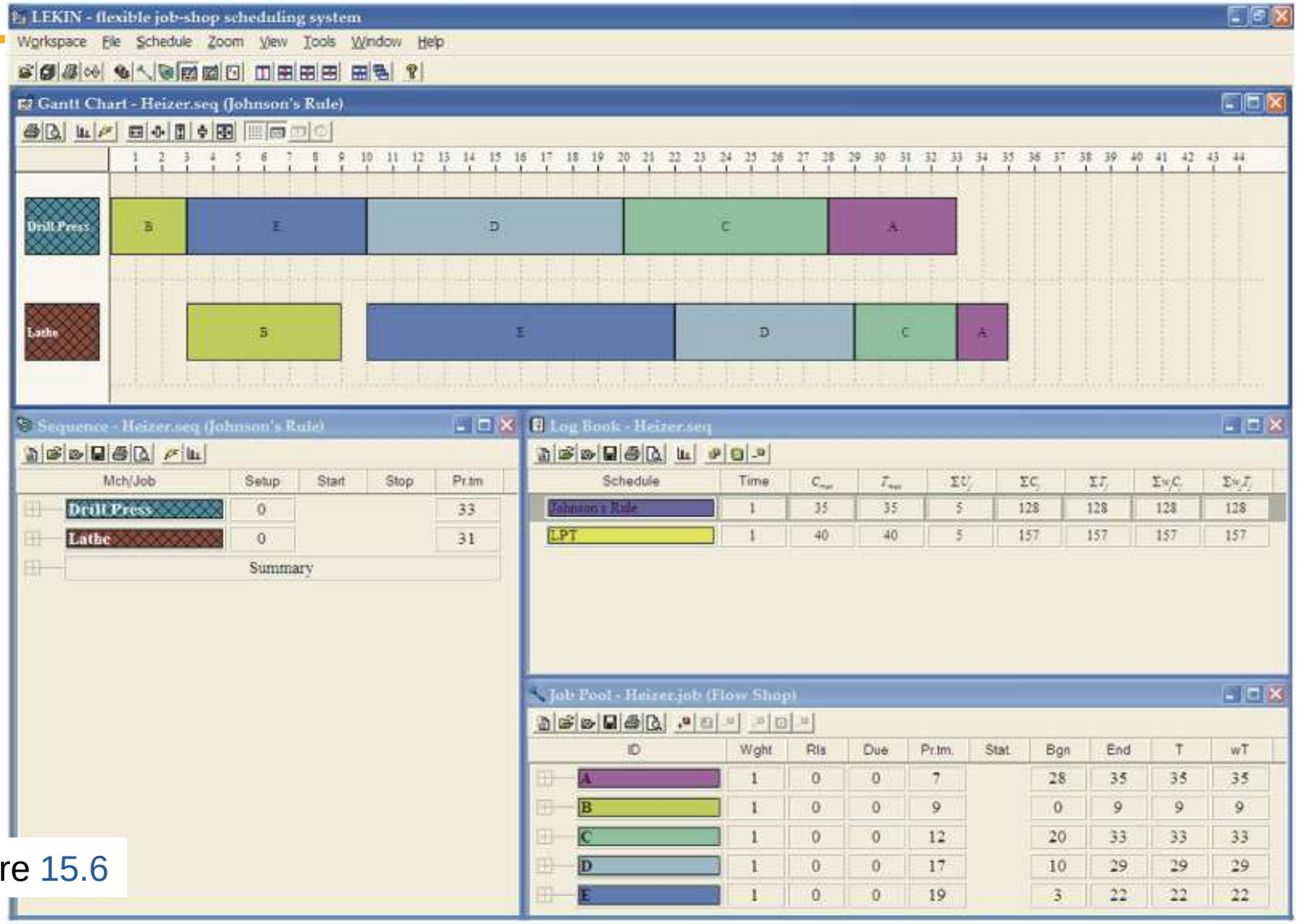


Figure 15.6

Scheduling Services

Scheduling Manufacturing	Scheduling Services
Emphasis on machines and materials	Emphasis on staffing levels
Maintain inventories	Do not maintain inventories
Not a labour intensive	Labour intensive with variable demand in labour
Schedule materials and machines	Schedule people

Scheduling Services

- Hospitals have complex scheduling systems to handle complex processes and material requirements
- Banks use a cross-trained and flexible workforce and part-time workers
- Retail stores use scheduling optimization systems that track sales, transactions, and customer traffic to create work schedules in less time and with improved customer satisfaction

Scheduling Services

- Airlines must meet complex FAA and union regulations and often use linear programming to develop optimal schedules
- 24/7 operations like police/fire departments, emergency hot lines, and mail order businesses use flexible workers and variable schedules, often created using computerized systems

Scheduling Service Employees With Cyclical Scheduling

- Objective is to meet staffing requirements with the minimum number of workers
- Schedules need to be smooth and keep personnel happy
- Many techniques exist from simple algorithms to complex linear programming solutions

Cyclical Scheduling

- Cyclical scheduling focuses on developing varying (inconsistent) schedules with the minimum number of workers.
- In these cases, each employee is assigned to a shift and has prescribed time off.
- **Take Home Assignment : Example 8 of Text book.**

Cyclical Scheduling

Example 8

CYCLICAL SCHEDULING

Hospital administrator Doris Laughlin wants to staff the oncology ward using a standard 5-day work-week with two consecutive days off, but also wants to minimize the staff. However, as in most hospitals, she faces an inconsistent demand. Weekends have low usage. Doctors tend to work early in the week, and patients peak on Wednesday then taper off.

APPROACH ► Doris must first establish staffing requirements. Then the following five-step process is applied.

SOLUTION ►

1. Doris has determined that the necessary daily staffing requirements are:

DAY	MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY	SUNDAY
Staff required	5	5	6	5	4	3	3

2. Identify the two consecutive days that have the *lowest total requirement* and circle these. Assign these two days off to the first employee. In this case, the first employee has Saturday and Sunday off because 3 plus 3 is the *lowest sum* of any 2 days. In the case of a tie, choose the days with the lowest adjacent requirement, or by first assigning Saturday and Sunday as an "off" day. If there are more than one, make an arbitrary decision.
3. We now have an employee working each of the uncircled days; therefore, make a new row for the next employee by subtracting 1 from the first row (because one day has been worked)—except for the circled days (which represent the days not worked) and any day that has a zero. That is, do not subtract from a circled day or a day that has a value of zero.
4. In the new row, identify the two consecutive days that have the lowest total requirement and circle them. Assign the next employee to the remaining days.
5. Repeat the process (Steps 3 and 4) until all staffing requirements are met.

	MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY	SUNDAY
Employee 1	5	5	6	5	4	3	3
Employee 2	4	4	5	4	3	3	3
Employee 3	3	3	4	3	2	3	3
Employee 4	2	2	3	2	2	3	2
Employee 5	1	1	2	2	2	2	1
Employee 6	1	1	1	1	1	1	0
Employee 7						1	
Capacity (measured in number of employees)	5	5	6	5	4	3	3
Excess capacity	0	0	0	0	0	1	0

Further Reading / References

1. Operation Management by Jay Heizer, Barry Render & Chuck Munson—Chapter 15
2. Operation Management by William J Stevenson—Chapter 16

THANK YOU